

Sequences and Their Limits Part I

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I Introduction

Definition. A sequence $\{a_n\}_{i=1}^{\infty}$ has a limit L if a_n can be made arbitrarily close to L by taking n large enough. If the limit L exists and is finite, we say that the sequence *converges* to L and write:

$$\lim_{n \rightarrow \infty} a_n = L$$

If the limit does not exist, or is infinite, we say that the sequence *diverges*.

Theorem. Let $y = f(x)$ be a function and $\{a_n\}_{i=1}^{\infty}$ be a sequence. If

$$\lim_{n \rightarrow \infty} f(x) = L \text{ and } f(n) = a_n \text{ when } n \text{ is a positive integer, then } \lim_{n \rightarrow \infty} a_n = L$$

Remark. The limit laws also apply to sequences.

Recall. We know that for $r > 0$,

$$\lim_{x \rightarrow \infty} \frac{1}{x^r} = 0$$

So, by our theorem, it follows that the sequence $\left\{ \frac{1}{n^r} \right\}_{i=1}^{\infty}$ also has a limit equal to 0.

Recall. If a is a positive real number, then

$$\lim_{x \rightarrow \infty} a^x = \infty \text{ if } a > 1, \lim_{x \rightarrow \infty} a^x = 0 \text{ if } 0 < a < 1$$

So, it follows that if $0 < a < 1$ then $\{a_n\}_{n=1}^{\infty}$ converges to 0 and if $a > 1$ the same sequence diverges.

[†]Definitions and example problems from MATH 12003: Analytic Geometry and Calculus II instructed by Dr. Donald White

Theorem. If $\{b_n\}_{n=1}^{\infty}$ is a sequence such that $\lim_{n \rightarrow \infty} |b_n| = 0$, then $\lim_{n \rightarrow \infty} b_n = 0$ (because the absolute values get closer to 0).

So, if $-1 < a < 1$ then $\lim_{n \rightarrow \infty} |a^n| = \lim_{n \rightarrow \infty} |a|^n = \lim_{n \rightarrow \infty} a^n = 0$.

Finally, note that if $a \leq -1$, the sequence diverges and if $-1 < a \leq 1$ the sequence converges.

II Examples

1. Find the limit of the sequence

$$\left\{ \frac{2n^3 - 3n^2 + 5}{7n^3 + n - 6} \right\}_{n=1}^{\infty}$$

Solution. A trick we can use to evaluate the limit of this sequence is to divide both the numerator and denominator by the highest power in the denominator. In this case its n^3 :

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{(2n^3 - 3n^2 + 5)^{\frac{1}{n^3}}}{(7n^3 + n - 6)^{\frac{1}{n^3}}} \\ = \lim_{n \rightarrow \infty} \frac{2 - \frac{3}{n} + \frac{5}{n^3}}{7 + \frac{1}{n^2} - \frac{6}{n^3}} \\ = \frac{2 - 0 + 0}{7 + 0 - 0} = \frac{2}{7} \end{aligned}$$

So, the sequence converges to $\frac{2}{7}$.

2. Does $\{2^n\}_{n=1}^{\infty}$ converge or diverge?

Solution. Since $a > 1$ the sequence diverges.

3. Does $\{\frac{1}{2^n}\}_{n=1}^{\infty}$ converge or diverge?

Solution. The sequence converges to 0 since $n > 0$.

4. Does $\{(-1)^n\}_{n=1}^{\infty}$ converge or diverge?

Solution. The sequence diverges since $a = -1$